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Youth Soccer Participation and Brain Health Outcomes in Adolescent Athletes

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Abstract

IMPORTANCE Repetitive head impacts (RHIs) are associated with later-life neurodegeneration. Because soccer is the most widely played sport among youth worldwide, identifying early changes associated with RHI is important.

OBJECTIVE To determine whether participation in 1 season of youth soccer is associated with changes in cognition, behavior, balance, brain structure or function, or blood biomarkers compared with noncontact sports.

DESIGN, SETTING, AND PARTICIPANTS Prospective longitudinal cohort study at European centers (Munich, Germany; Leuven, Belgium; and Oslo, Norway). Male adolescent soccer players and noncontact athletes were each studied across a single competitive season with assessments at preseason, postseason, and 2 months later. Data were analyzed from January 2023 to March 2025.

EXPOSURES Soccer players were compared with noncontact athletes. In addition, self-reported heading of a soccer ball was assessed among soccer players as a measure of RHI.

MAIN OUTCOMES AND MEASURES Cognition, behavior, balance, magnetic resonance imaging (brain structure, function, and biochemistry), and plasma biomarkers.

RESULTS Male adolescent soccer players (n = 82; mean [SD] age, 14.8 [0.6] years) did not differ from noncontact sport athletes (n = 47; mean [SD] age, 14.7 [0.7] years) in cognition, behavior, balance, cortical thickness, brain volumes, white-matter microstructure, or functional connectivity. At preseason, soccer players had higher total N-acetylaspartate (tNAA; β, −0.379 [95% CI, −0.627 to −0.131]; *P* = .003), glial fibrillary acidic protein (GFAP; β, −0.055 [95% CI, −0.103 to −0.006]; *P* = .03), and neurofilament light chain (NfL; β, −0.071 [95% CI, −0.122 to −0.020]; *P* = .01) than noncontact sport controls. Across the season, tNAA (β, 0.047 [95% CI, 0.020-0.074]; *P* = .001) declined in soccer players and increased in controls, converging by postseason. Group trajectories of GFAP and NfL did not differ between groups. Within soccer players, heading exposure was not significantly associated with changes in any outcome.

CONCLUSIONS AND RELEVANCE In this cohort study of adolescent males, no statistically significant differences were detected over 1 season between soccer players and noncontact sport athletes in cognition, behavior, or brain structure and function. Group differences in GFAP and NfL may represent early signs of exposure, but lack of association with heading exposure warrants further investigation. These results highlight the need for large, multiyear studies to inform health policy.

Key Points

Question Is participation in youth soccer associated with short-term changes in cognition, behavior, brain structure and function, or biomarkers of brain injury compared with noncontact sports?

Findings In this cohort study of 129 youth athletes, no statistically significant differences were detected over 1 season between soccer players and noncontact athletes in cognition, behavior, balance, neuroimaging metrics, or blood biomarkers.

Meaning Although this study found no difference in brain-related outcomes over 1 season between youth soccer players and noncontact sports athletes, large, multiyear studies are required to inform health policy.

+ Supplemental content

Author affiliations and article information are listed at the end of this article.

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Introduction

Repetitive head impacts (RHIs), defined as repeated impacts to the head that do not cause clinical concussion, have been associated with long-term neurologic sequelae, including chronic traumatic encephalopathy and other neurodegenerative diseases.[1-3] Soccer is the world's most widely played sport, with more than 22 million adolescents participating globally.[4] At the same time, concerns have emerged about the safety of heading a soccer ball—a unique aspect of soccer that introduces RHI exposure—during a period of rapid brain maturation.[5,6] This overlap has fueled international debate and led to precautionary policies, yet scientific evidence to guide such policies remains limited.

Epidemiological studies of retired professional soccer players have reported elevated risks of neurodegenerative disease,[2,3] and experimental studies reported transient changes in brain structure and function following controlled heading.[7-11] Yet evidence in adolescents remains scarce. Existing studies are typically small, cross-sectional, and heterogeneous in design,[1,12] leaving uncertainty about whether routine participation in youth soccer has measurable short-term effects on cognition, behavior, neuroimaging measures, or biomarkers of brain injury.

This uncertainty has driven parental concern, media debate, and precautionary policies, including heading restrictions in youth soccer in some countries. Evidence from well-powered, prospective, longitudinal studies is critical to inform recommendations.

The REPIMPACT study was designed to address this need.[13] Using harmonized protocols across 3 European centers, we assessed cognition, behavior, balance, multimodal magnetic resonance imaging (MRI) (structure, diffusion, and resting-state functional MRI), magnetic resonance spectroscopy, and plasma biomarkers across a single competitive season in adolescent male soccer players and male noncontact sport athletes. Given the heterogeneity of prior findings and uncertainty regarding which domains are most sensitive to short-term RHI exposure in adolescents, a multimodal approach was used to assess a range of previously associated outcomes within a single, standardized design. As prespecified in the previously published study protocol,[13] the primary analysis evaluated between-group differences in change over time. Secondary analyses included testing for preseason group differences and, within soccer players, associations between heading exposure and changes in outcomes.

Methods

Study Design and Setting

The REPIMPACT study was a prospective, longitudinal, multicenter cohort study of adolescent athletes conducted at 3 European sites (Munich, Germany; Leuven, Belgium; and Oslo, Norway) with repeated assessments at 3 time points over approximately 1 competitive season (time point 1, preseason; time point 2, postseason; and time point 3, 2 months postseason) (eAppendix 1 in Supplement 1). Data acquisition was performed from July 2017 to May 2020, with analyses from January 2023 to March 2025. The study design as well as detailed sample characteristics have previously been described.[13] Given the exploratory, multimodal nature of the study and the limited prior data to inform expected effect sizes across outcome domains in adolescent populations, a formal a priori sample size or power calculation was not performed. The sample size was determined based on feasibility across participating sites and was intended to enable detection of moderate group-level changes over time within a harmonized longitudinal framework. The study was approved by the ethics committees at all participating sites. Written informed consent was obtained from all participants and their legal guardians prior to participation, in accordance with the Declaration of Helsinki.[14] This study is reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline.

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Participants

We enrolled male soccer players between 14 and 16 years of age and male noncontact sport athletes group-matched by age. Soccer players were recruited from academy-level organized clubs with regular training schedules (≥3 sessions per week). Control athletes participated in competitive noncontact sports (eg, swimming, athletics, cycling, and racket sports) and, similar to soccer players, trained at least 3 times per week. Exclusion criteria included prior physician-diagnosed concussion or traumatic brain injury, neurologic or psychiatric disorders, learning disabilities, or MRI contraindications as prespecified in the study protocol.[15] Participants with a history of physician-diagnosed concussion were excluded to reduce baseline heterogeneity and to better isolate associations of RHI-related exposure with soccer participation during the study period.

Exposure

Heading exposure was assessed using a structured questionnaire capturing the mean number of headers per training session and per match, along with the frequency of sessions per week, allowing estimation of typical weekly and seasonal exposure. Self-reported values were collected at preseason and postseason, and the mean was calculated to derive a season-level estimate (details in eAppendix 2 in Supplement 1). Self-report was used because in-ear sensor data, while tested, showed high random errors and overestimation in adolescents, limiting feasibility.[15,16] These estimates should therefore be interpreted as relative rather than absolute measures of exposure.

Assessments

Participants were examined at 3 time points. Cognition and behavior were assessed using computerized neurocognitive testing (Cogstate), Youth Self-Report, and standardized fluid intelligence testing (Cattell Culture Fair Test 20-R) (details in eAppendix 3 and eTable 1 in Supplement 1). Balance was assessed using the center-of-pressure sway path length on a computerized balance board (details in eAppendix 4 in Supplement 1.[17-19]Neuroimaging was conducted using a harmonized MRI protocol across sites and including structural MRI, diffusion MRI, resting-state functional MRI, and magnetic resonance spectroscopy focused on periventricular white matter N-acetylaspartate (tNAA).[20-25] The MRI sequence parameters are provided in eTable 2 in Supplement 1. Finally, we assessed the concentrations of 2 fluid biomarkers, plasma neurofilament light chain (NfL) and glial fibrillary acidic protein (GFAP), as measured with ultrasensitive assays.[26-28] Details of MRI acquisition protocols, quality control, and micro-RNA (miRNA) analyses are provided in eAppendix 5 and 6 and eTable 2 in Supplement 1.

Statistical Analysis

Linear mixed-effects models were used to evaluate between-group differences in longitudinal change across time points (time point 1 minus time point 2; time point 2 minus time point 3), adjusting for age, geographic site, and time interval between visits. As prespecified in the study protocol,[15] this analysis represented the primary comparison. Secondary analyses included assessment of preseason (time point 1) group differences and, within soccer players, associations between heading exposure and changes in outcomes using linear regression. Group differences across tertiles of heading exposure were also explored (eTable 3 in Supplement 1). Results were corrected for multiple comparisons using the false discovery rate,[29] with a 2-sided $P < .05$ considered statistically significant. All analyses were conducted using SAS Studio, version 3.82 (Enterprise Edition; SAS Institute Inc), and SPSS, version 29.0.2.0 (IBM) (details in eAppendix 7 in Supplement 1).

Results

Participants

We enrolled 129 male athletes: 82 soccer players (mean [SD] age, 14.8 [0.6] years) and 47 noncontact sport athletes (mean [SD] age, 14.7 [0.7] years). The 2 groups were similar in age, body mass index,

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and training load (**Table**). Of 129 enrolled athletes, 123 completed preseason, 115 completed postseason, and 108 completed the 2-month postseason follow-up assessments (Table). Reasons for missing visits included logistical constraints and pandemic-related disruptions at specific sites. Heading exposure varied substantially among soccer players (mean [SD], 939 [696] headers per season), indicating considerable heterogeneity in exposure levels (eAppendix 2 in Supplement 1).

Behavior, Cognition, and Balance

No significant group differences were observed at preseason and no significant group differences were observed across time (all data given here are examples of analyses on group differences in slope between time points 1 and 2) in internalizing (β, 0.007 [95% CI, −0.272 to 0.286]; *P* = .96) or externalizing (β, −0.042 [95% CI, −0.286 to 0.203]; *P* = .96) problems, total problem scores (β, −0.015 [95% CI, −0.229 to 0.200]; *P* = .96), executive function (β, 0.004 [95% CI, −0.025 to 0.033]; *P* = .97), memory (β, −0.001 [95% CI, −0.033 to 0.032]; *P* = .97), processing speed (β, −0.013 [95% CI, −0.042 to 0.016]; *P* = .97), or balance (β, −0.192 [95% CI, −0.487 to 0.102]; *P* = .20)

Table. Characteristics of Soccer Players and Noncontact Sport Athlete Controls

Characteristic	Participants, No. or mean (SD)		*P* value
	Soccer	Controls	
Sample size, No.			
Preseason	82	47	NA
Postseason	74	41	NA
2 mo postseason	68	40	NA
Site distribution, No.			
Preseason			
Leuven, Belgium	20	13	.08
Munich, Germany	31	9	
Oslo, Norway	31	25	
Postseason			
Leuven, Belgium	19	12	.06
Munich, Germany	26	6	
Oslo, Norway	29	23	
2 mo postseason			
Leuven, Belgium	18	11	.16
Munich, Germany	21	6	
Oslo, Norway	29	23	
Days between time points			
Preseason-postseason	287.9 (32.9)	292.3 (27.1)	.47
Postseason–2 mo postseason	86.4 (31.9)	72.0 (24.1)	.01
Age, mean (SD), y			
Preseason	14.8 (0.6)	14.7 (0.7)	.57
Postseason	15.6 (0.6)	15.5 (0.7)	.50
2 mo postseason	15.7 (0.5)	15.7 (0.7)	.64
School grade level (international scale)			
Preseason	9.1 (0.7)	9.3 (0.8)	.33
Postseason	9.7 (0.7)	10.0 (0.8)	.14
2 mo postseason	10.2 (0.6)	10.4 (1.0)	.28
Body mass index[a]			
Preseason	19.6 (1.8)	20.0 (1.6)	.19
Postseason	20.4 (1.9)	20.6 (1.7)	.67
2 mo postseason	20.5 (1.8)	21.0 (2.0)	.27
Years of soccer play at preseason	9.1 (1.7)	NA	NA
Estimate of RHI exposure over the season as No. of headers	939 (696)	NA	NA

Abbreviations: NA, not applicable; RHI, repetitive head impact.

[a] Body mass index calculated as weight in kilograms divided by height in meters squared.

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(all false discovery rate corrected) (eTable 5, eFigure 1, and eFigure 2 in Supplement 1). Among soccer players, heading exposure was not associated with changes in these measures. Full statistical results for all measures can be found in eTable 5 in Supplement 1.

Neuroimaging

Cortical thickness (eg, left hemisphere β, −0.001 [95% CI, −0.004 to 0.002]; *P* = .48), total gray (β, 0.000 [95% CI, −0.001 to 0.001]; *P* = .38) and white (β, 0.000 [95% CI, 0.000-0.000]; *P* = .38) matter volumes, diffusion-derived white matter measures (FA β, 0.000 [95% CI, −0.001 to 0.000]; *P* = .90), and resting-state default mode network connectivity (β, 0.000 [95% CI, −0.003 to 0.003]; *P* = .98) did not differ between groups longitudinally (all data reported here are examples of analyses on group differences in slope between time points 1 and 2). Similarly, these neuroimaging measures did not differ between groups at preseason (eFigure 3, and eFigure 4 in Supplement 1). In addition, no associations with heading exposure were detected.

Magnetic Resonance Spectroscopy

At preseason, soccer players had a higher periventricular white matter tNAA concentration than noncontact sport controls (β, −0.379 [95% CI, −0.627 to −0.131]; *P* = .003). Over the season, the values converged, with tNAA decreased in soccer players and increased in controls, and with a significant group × time interaction (β, 0.047 [95% CI, 0.020-0.074]; *P* = .001) (**Figure 1**). Postseason values remained stable. No associations with heading exposure were detected.

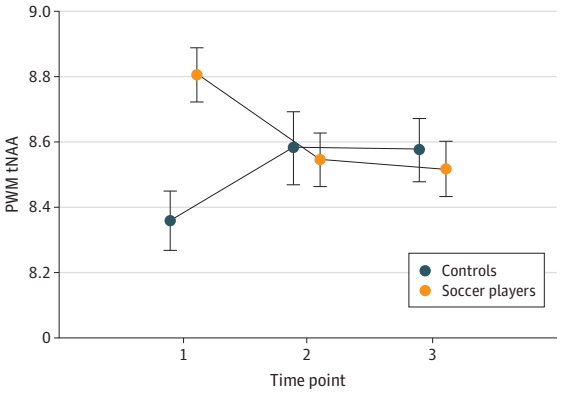
Fluid Biomarkers

At preseason, soccer players had higher GFAP (β, −0.055 [95% CI, −0.103 to −0.006]; *P* = .03) and NfL (β, −0.071 [95% CI, −0.122 to −0.020]; *P* = .01) concentrations compared with controls. Over the season, GFAP decreased (β, 0.000 [95% CI, −0.006 to 0.007]; *P* = .90) and NfL increased (β, 0.000 [95% CI, −0.005 to 0.006]; *P* = .90) in both groups, with no significant between-group differences in rate of change (**Figure 2**). No associations with heading exposure were detected.

miRNA

No group differences were observed at any time point (eg, time point 1 controls vs time point 1 soccer, hsa-miR-145-5p, *P* = .88). In addition, no associations with heading exposure were identified (eTable 4 in Supplement 1).

Figure 1. Point-Estimate Plot Illustrating Group Comparisons in Brain Biochemistry



PWM indicates periventricular white matter; tNAA, total N-acetylaspartate concentration. Error bars represent SE. Time point 1 is preseason; 2, postseason; and 3, 2 months postseason. Significant findings were *P* = .003 between groups at time point 1, and *P* = .001 for the group × time interaction.

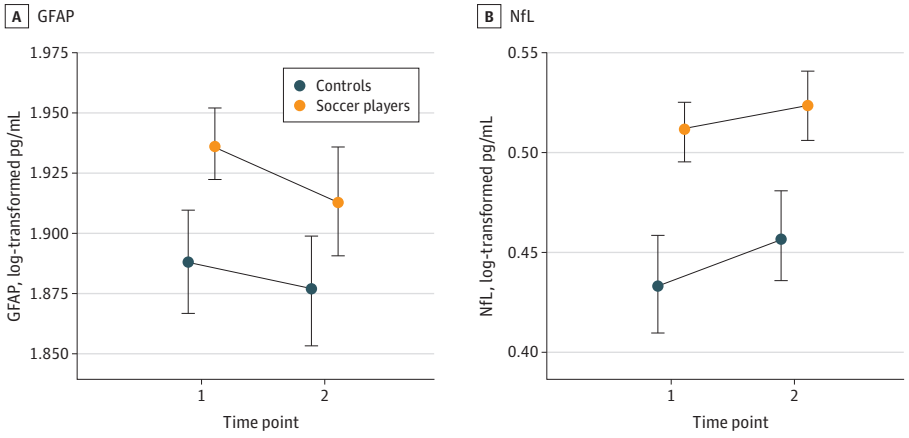
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Discussion

In this multinational, prospective cohort of adolescent athletes, no statistically significant between-group differences were detected in cognition, behavior, balance, cortical thickness, brain volumes, white-matter microstructure, or functional connectivity when comparing soccer players with age-matched noncontact sport athletes over a single season. The absence of between-group differences across imaging, cognitive, and behavioral domains contrasts with some smaller studies in youth soccer.[1,7-9] Possible explanations include our larger sample, harmonized multicenter protocol, and rigorous quality control, which may reduce measurement variability. It is also possible that neurobiological changes associated with RHI in adolescence are transient or too subtle to detect over a single season. The absence of group differences and lack of associations with self-reported heading exposure should be interpreted cautiously. Importantly, these findings reflect the additional measurable differences associated with 1 competitive season in athletes with several years of prior soccer participation, rather than the absence of any changes associated with soccer exposure per se. Accordingly, the results indicate that no additional neurobiological change associated with soccer was detectable over the study period within the sensitivity limits of this sample. Moreover, lack of statistically significant group differences over a single season does not exclude any long-term effects, because neuropathological processes may require years of cumulative exposure.

Our secondary hypothesis was partly supported: soccer players showed higher preseason levels of tNAA, NfL, and GFAP than controls. By postseason, tNAA values converged between groups, suggesting a transient difference without clear evidence of association with RHI exposure. In contrast, GFAP and NfL trajectories remained parallel, with higher mean values in soccer players throughout the season. Baseline assessments were conducted during the preseason and may have occurred in close temporal proximity to recent training; these elevations may therefore reflect short-term physiological responses rather than cumulative residual findings associated with prior exposure. Alternatively, they may represent adaptive processes, residual responses associated with prior exposure. Many of the soccer players in our study had a long history of participation in soccer, which may have contributed to preseason differences and may reflect cumulative exposure not fully captured by single-season estimates.[26,27] The levels of GFAP and NfL biomarkers are within the normal physiological range for this age[30] and consistent with the reported effects of RHI exposure in contact sports.[28] Given the modest sample size and the exploratory nature of these biomarker findings, they should be interpreted with caution. Longer-term longitudinal studies across multiple seasons, with more precise characterization of recent and cumulative exposure, are needed to determine whether these biomarker patterns are clinically meaningful and relate to the increased prevalence of neurodegeneration reported in former professional players.

Figure 2. Point-Estimate Plots Showing Fluid Biomarker Concentrations for Glial Fibrillary Acidic Protein (GFAP) and Neurofilament Light (NfL)



Time point 1 indicates preseason; 2, postseason. Error bars represent SE. Statistically significant findings were GFAP concentrations between groups at time point 1 (P = .03) and NfL concentrations between groups at time point 1 (P = .01).

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Limitations

Limitations include reliance on self-reported heading exposure, which may have introduced misclassification due to recall inaccuracy and assumptions of stable heading frequency across the season, potentially biasing exposure–outcome associations toward the null and reducing sensitivity to detect dose–response relationships. Substantial heterogeneity in heading exposure within the soccer group was not fully captured, and group-level comparisons may obscure alterations restricted to individuals with higher exposure. Objective impact quantification was not feasible in this age group due to limitations of in-ear sensors, and self-reported estimates are not directly comparable to objectively measured head impacts.

As this was an observational, nonrandomized study, residual confounding cannot be excluded, and unmeasured differences between groups may influence the findings. Although both groups were competitively active and trained regularly, detailed and comparable measures of cumulative training load were not systematically captured. While self-reported information on average training frequency and match exposure was available, objective or exact counts of training sessions and games were not collected, which may limit precision in estimating overall exposure.

The study was restricted to adolescent male athletes in European academy settings and may not generalize to female athletes, younger children, or athletes in other geographic or training contexts. In addition, exclusion of participants with prior physician-diagnosed concussion may further limit generalizability and precludes evaluation of how prior injury modifies susceptibility to RHI.

The study may have been underpowered to detect small effects, and formal estimates of detectable effect sizes were not derived; therefore, absence of statistical significance should be interpreted as absence of detected differences within the sensitivity limits of this study rather than absence of effect. Finally, the multimodal design did not prioritize a single primary outcome domain, which may reduce sensitivity to detect domain-specific changes and complicate interpretation across measures.

Conclusions

In this cohort study of adolescent soccer players compared with noncontact athletes, no statistically significant group differences were detected in cognition, behavior, or brain structure and function across a single competitive season. At preseason, soccer players had higher tNAA, GFAP, and NfL; tNAA declined in players and increased in controls, converging by postseason, and GFAP and NfL trajectories did not differ between groups. In the absence of exposure-response associations, these preseason biomarker differences may reflect recent exposure, training-related physiology, or other unmeasured factors not specific to RHIs. Overall, no statistically significant short-term differences were detected over 1 season. Risk mitigation approaches such as age-appropriate heading guidance, education, and large, multiyear studies remain essential to further inform child health policy.

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Author Contributions: Drs Koerte and Wiegand had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Drs Koerte and Wiegand contributed equally to this work and are considered co–first authors.

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Critical review of the manuscript for important intellectual content: All authors.

Statistical analysis: Wiegand, Seitz-Holland, Tripodis.

Obtained funding: Koerte, Cente, Shenton, Bahr, Filipcik, Gooijers, Swinnen, Pasternak.

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Supervision: Koerte, Bahr, Filipcik, Gooijers, Leemans, Pasternak.

Conflict of Interest Disclosures: Dr Koerte reported receiving grants from ERA-NET Neuron, the European Research Council, and the National Institutes of Health during the conduct of the study; having a spouse who is member of Siemens managing board outside the submitted work; and being President of the European Neurotrauma Organisation (unpaid position). Dr Seitz-Holland reported receiving grants from the National Institute of Mental Health, Harvard Medical School, the Brain and Behavior Research Foundation, and the Women's Brain Initiative during the conduct of the study. Dr Cente reported receiving grants from the Slovak Research and Development Agency and the Ministry of Education, Research, Development and Youth of the Slovak Republic during the conduct of the study. Dr E. Kaufmann reported receiving grants from Deutsche Forschungsgemeinschaft and Medical Clinician Scientist Program at the Medical Faculty at LMU; and receiving personal fees from UCB, Neuraxpharm, Desitin, Eisai, UNEEG, and Precisis outside the submitted work. Dr Bonfert reported receiving grants from the Federal Joint Committee (G-BA), the ZNS Foundation, and from CSL Behring GmbH; and receiving personal fees from Infectopharm, TEVA, and Zurich insurance outside the submitted work. Dr Lin reported being a co-founder of Brainspec Inc; receiving grants from Biomarin Pharmaceuticals; and receiving personal fees from Moncton outside the submitted work. Dr Tripodis reported receiving personal fees

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Data Sharing Statement: See Supplement 3.

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SUPPLEMENT 1.

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SUPPLEMENT 2.

REPIMPACT Consortium Investigators

SUPPLEMENT 3.

Data Sharing Statement

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